

Nikhil Singhal

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SKILLS

Programming Languages: C++ (3+ years), Python (3+ years), JavaScript/TypeScript (3+ years), Rust (1+ years)

Computer Vision & Deep Learning: Instance Segmentation, Object Detection, 6-DOF Pose Estimation, Point Cloud Processing, Depth Sensing, Transformer-based Vision (DETR), CNN Architectures, Sim-to-Real Transfer, Synthetic Data Generation, PyTorch, TensorFlow, OpenCV, RealSense SDK

Robotics: ROS2, MoveIt2, Nav2, SLAM, Motion Planning, Behavior Trees, Gazebo, MuJoCo, PyBullet, Isaac Sim, OpenSCENARIO 2, PDDL

Systems & Deployment: Docker, Kubernetes, Helm, AWS, Edge Computing, Microservices, gRPC, ZMQ

Development Tools: Git/GitHub, CI/CD Pipelines, Automated Testing, Ansible, Jira

Languages: English (C1), German (B2), Hindi (Native)

EXPERIENCE

Robotics Software Engineer

Unchained Robotics GmbH

August 2025 - Present, Paderborn, Germany

- Owned end-to-end development of a deep-learning perception pipeline for 6-DOF pose estimation of mixed, occluded items, integrating industrial 3D cameras, benchmarking DETR, YOLO, and Mask R-CNN, and selecting a DETR-based instance-segmentation model for its stronger occlusion handling and lower fine-tuning-data requirements.
- Designed and deployed a classical computer-vision pipeline from scratch (OpenCV: contour detection, morphological operations, plane fitting, geometric validation) on Intel RealSense RGB-D data, achieving pose accuracy within 5 mm translation and 1 degree rotation, under 500 ms capture-to-pose latency.
- Implemented eye-in-hand and eye-to-hand camera calibration to align 3D camera data with robot coordinate frames, enabling accurate transfer of estimated poses to robot motion.
- Deployed the perception system into continuous 24/7 production, processing roughly 1,500 items per day while handling extensive edge cases to meet strict reliability and cycle-time targets.
- Built physics-based synthetic data generation in NVIDIA Isaac Sim, validated through the identical production pipeline, improving model generalization via sim-to-real transfer and cutting reliance on manual labeling.
- Enhanced the core Python framework serving as the software PLC for full robotic-cell orchestration, improving state-machine logic (topological sort for execution ordering), concurrency (threading and multiprocessing), and FastAPI service interfaces.
- Contributed to migrating inter-component communication to a gRPC-based architecture while orchestrating multiple robot arms (KUKA, Nachi, UR) across large-scale production deployments.

Robotics Software Developer and Researcher (Internship and Master Thesis)

Intel Corporation (Intel Labs)

January 2024 - January 2025, Karlsruhe, Germany

- Contributed to the open-source Scenario Execution library (OpenSCENARIO DSL and Behavior Trees) for reproducible robotics testing, working across the full ROS2 stack in C++ and Python (topics, services, actions, rosbags) and integrating Nav2 navigation with MoveIt2 manipulation in Gazebo simulation.
- Built CI/CD pipelines with GitHub Actions and automated unit testing, improving deployment efficiency and speeding delivery of updates to the library.
- Designed a Kubernetes-based fault-tolerance framework (master thesis) enabling a mobile manipulation system to recover automatically from mid-task component failures, with no external intervention, in around 1.5 seconds.
- Implemented stateful recovery, restoring the system to its saved pre-failure state rather than restarting the task from scratch, and packaged deployments as Helm charts for versioned, repeatable rollouts across the cluster, substantially reducing downtime in edge-computing environments.
- Offloaded resource-intensive ROS2 workloads (MoveIt2, Nav2) from onboard hardware to a remote server over real-time communication, reducing onboard resource usage by 80%.
- Validated recovery strategies in Gazebo and on physical robots, analyzing performance metrics to refine fault-recovery approaches for reliable operation.

Robotics Software Developer and Researcher (Student Research Assistant)

Institute of Mobile Machines and Commercial Vehicles, TU Braunschweig

March 2023 - December 2023, Braunschweig, Germany

- Developed a ROS2 framework for sensor-data exchange across a heterogeneous multi-machine system (drone, ground robot, edge device, cloud server), letting each machine process or store data captured by another, for example streaming drone imagery to a ground robot to guide its navigation.
- Implemented image-chunking for reliable transfer of large TIFF captures over a constrained 5G network, eliminating single-payload failures and sustaining high throughput on a lossy connection.
- Configured the network interfaces so all machines discover one another under a single ROS2 network, a prerequisite for cross-machine ROS2 communication.

EDUCATION

M.Sc. Computational Sciences in Engineering

TU Braunschweig • Braunschweig, Germany • 2025

- Relevant Courses:** Robotics, Robot Control and Optimization, Pattern Recognition, Advanced Programming, Computer Vision and Machine Learning.

PUBLICATIONS

- IEEE ICRA 2024 - "A Generic Approach for Reactive Stateful Mitigation of Application Failures in Distributed Robotics Systems Deployed with Kubernetes." Mirus F., Pasch F., Singhal N., Scholl K.-U. (CoRR) 2024 [URL]